



Supplier intelligence for **semiconductor** **procurement**

Semitrical turns a semiconductor supplier's own public filings into procurement intelligence, so a small buying team walks into a negotiation with the depth the supplier's commercial team already has.

SUPPLIER ANALYSIS · MEMORY AND ANALOG ·
UPDATED AS THEY FILE

What the Roadmap Leaves Out

What eleven memory and analog suppliers' own filings say about capacity, investment and staying power — read from the buyer's side.

WHAT THIS ANALYSIS COVERS — ELEVEN SUPPLIERS, BOTH
GROUPS

MEMORY

Micron · SK hynix · Samsung · Kioxia · Nanya · Winbond

ANALOG

Texas Instruments · STMicroelectronics · Infineon · NXP · Analog Devices

PREPARED FOR

Teams that buy and qualify semiconductors for vehicle programmes — component engineering, procurement and programme finance

SOURCES

Public company filings only. Nothing forecast, nothing modelled.

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POSITION AS AT

September 2026. Each supplier's figures change when it next files — see Method and sources.

CIRCULATION

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A roadmap says what a supplier intends to build. The filings say whether it can.

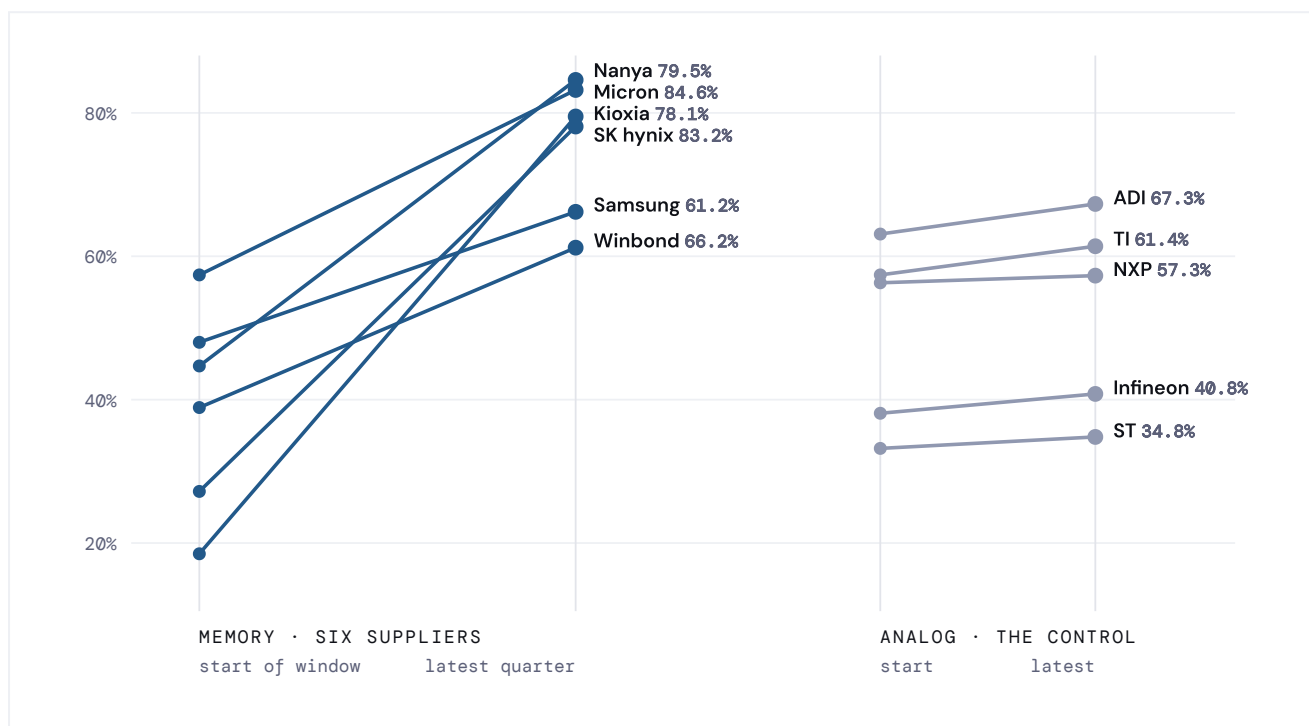
Only the first is ever presented. This analysis reads the second, for the eleven memory and analog suppliers behind a vehicle programme.

PRESENTED IN EVERY SUPPLIER REVIEW The roadmap Nodes, densities, capacity, product generations. What they intend to build.		RARELY OPENED The filings Cash, plant, investment direction. Whether the money exists to build it.	
	01 A price cycle is running, and it is visible Six memory makers moved together, up to 84.6% gross margin. All five analog suppliers did not. the cycle		02 Memory's implied price rose between ×1.5 and ×4 Analog over the same quarters: ×1.02 to ×1.13. the price move
	03 The eleven are in very different positions Three at their six-year peak. One 30% below, same quarter. the eleven		04 Some capacity is funded by the cycle, some by the business SK hynix ×2.54 price on 2.17× capex. TI ×1.10 on 2.37×. price move · fifteen years
	05 Two have cut plant spending below replacement Winbond 0.53×, about half. NXP 0.71×, its lowest in six years. fifteen years		06 The scale of the industry's capital problem \$158bn of revenue produced \$1.3bn of free cash in six years. the capital problem

Every figure comes from a document the company filed itself. Nothing is forecast or modelled, and the limits section sets out plainly what this method cannot see.

Six memory makers moved together. Analog did not move at all.

Gross margin is the cleanest single line for this, because it strips out everything except what a company sells for against what it costs to make. Over the four most recent reported quarters, every memory supplier's gross margin rose — several of them enormously — while all five analog suppliers barely moved.



— Memory • six suppliers — Analog suppliers — the control group

Gross margin at the start of the four-quarter window against the latest reported quarter. Both panels share one scale. Only the two endpoints are plotted; the line between them is a connector, not a series of quarterly values.

The mechanism is the reason this matters. Contract prices moved; the cost of making a bit did not. Revenue therefore rose against a flat cost of revenue, and gross margin expanded hard. That is a price event, not a volume event or an efficiency gain, and the analog panel is what proves it — the same quarters, the same economy, the same customers, and almost no movement.

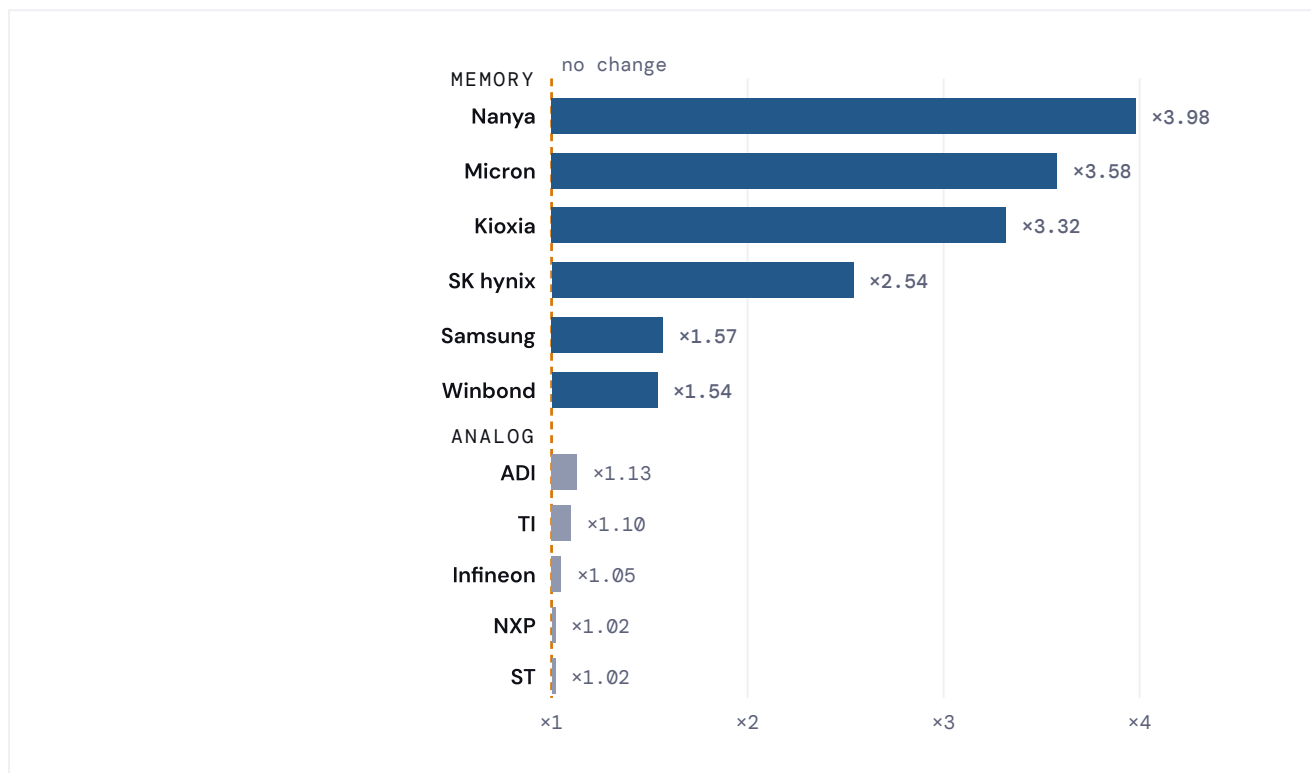
Micron is the sharpest illustration: gross margin reached 84.6% in the latest quarter, from 44.7% four quarters earlier. A company does not achieve that by manufacturing better.

TAKEAWAY

On its own this is an observation, not yet a finding. A margin chart says what happened to the sellers; it does not say what a buyer should do about it. It starts to mean something two pages on, when the same move is set against how much each supplier is spending on plant — because an identical price rise is paying for a new fab at one company and simply banking cash at another. **The margin is the fact. The build is what turns it into a read.**

Memory's implied price rose between ×1.5 and ×4. Analog did not move.

Gross margin is price minus cost, over price — so if unit costs held, the margins in the cycle chart say what happened to the price. **Cost of revenue barely moved across the window while revenue rose sharply, which is the evidence for holding cost still. The size of the move is not the size of the margin gain.** SK hynix added 25.8 margin points and Samsung 22.3 — nearly the same — for ×2.54 against ×1.57, because hynix started the window at a much higher margin.



Implied price multiple = $(1 - \text{gross margin at the start}) \div (1 - \text{gross margin now})$, from the same reported figures as the cycle chart. Bars measured from ×1.00. All eleven suppliers are shown; every starting margin in the current window is positive, so no row is excluded.

Blended, not a part number

Gross margin covers each company's whole product mix, so read these as the scale of the move rather than the price of one device.

The assumption biases upward

It holds unit costs unchanged. In an upcycle fabs run fuller, which spreads fixed cost over more units and lowers unit cost — so the true price move is likely smaller than shown. Read these as a ceiling.

What it separates

Put the price move next to what each supplier is spending on plant, and it answers a more useful question: **what is paying for the capacity being built?**

UNDERWRITTEN BY THE CYCLE

SK hynix

implied price capex ÷ dep

×2.54 **2.17×**

The largest build in the group, funded at a price level roughly double where it started. The plan depends on prices staying up.

UNDERWRITTEN BY THE BUSINESS

Texas Instruments

implied price capex ÷ dep

×1.10 **2.37×**

Heavier spending still, on a price that barely moved — and sustained through the downturn rather than started in the upswing.

TAKEAWAY

For a part that has to remain buyable for fifteen years, this is the distinction that matters: which capacity commitments are underwritten by a cycle, and which are underwritten by the business. It changes who is worth pressing for a longevity commitment in writing, and whose roadmap can be taken at face value. The fifteen-year question puts all eleven suppliers on the capital-spending side of it.

Eleven suppliers, two groups, one set of measures.

A vehicle programme buys memory and analog from a small number of companies. All eleven are shown on the same three measures, taken from the same filings: revenue against the highest of the last six years, capital spending as a multiple of depreciation, and free cash as a share of revenue. **Every figure below is each company's FY2025 — its latest reported full year.**

FY2025 for all eleven, but their fiscal years do not end on the same date, so this is each company at its own latest year-end rather than a single common quarter. Capital intensity is comparable only within a similar manufacturing model — the section on fab economics sets out how to read it. **Capital spending is measured against depreciation of plant, not total depreciation and amortisation** — plant depreciation is not present in our data for Kioxia or Samsung in any of the six years, so the ratio is not computed for them.

SUPPLIER	GROUP	REV VS 6-YR PEAK	CAPEX ÷ DEP FY25	FREE CASH MARGIN FY25
Micron MU	MEMORY	At peak	1.92×	4.5%
SK hynix 000660.KS	MEMORY	At peak	2.17×	26.6%
Samsung 005930.KS	MEMORY	-3.9%	not reported	11.3%
Kioxia 285A.T	MEMORY	At peak	not reported	14.2%
Nanya 2408.TW	MEMORY	-30.3%	0.96×	7.7%
Winbond 2344.TW	MEMORY	-19.5%	0.53×	5.2%
Texas Instruments TXN	ANALOG	-11.7%	2.37×	14.7%
STMicroelectronics STM	ANALOG	-31.7%	1.05×	2.6%
Infineon IFX	ANALOG	-6.2%	1.19×	9.4%
NXP NXPI	ANALOG	-7.6%	0.71×	19.7%
Analog Devices ADI	ANALOG	-10.5%	1.31×	38.8%

TAKEAWAY

A second source in the same category is not automatically a second position. Winbond and Nanya are both Taiwanese, both below their peak, both spending at or under replacement — a pair that looks like diversification but shares a country, a cycle and a plant posture. Meanwhile three memory companies sit at record revenue in the same quarter a fourth sits 30% below its own. **The category predicts almost nothing; the figures do.**

Each sparkline is **free cash flow margin** — cash from operations less capital spending, as a share of revenue — for 2020 to the latest full year. One shared scale across all eleven suppliers, -70% to +40%; the dashed line is zero, the marked point is the latest year.

Micron MU · memory

Rev vs peak Capex ÷ plant dep
At peak 1.92×

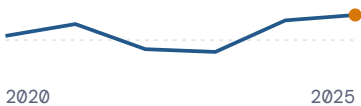
At its six-year revenue peak and spending nearly twice what its plant depreciates. Free cash margin never exceeded 10.1% in the period, including in the strongest years, because the cash goes into fabs. The 2023 trough reached -39.4%.



SK hynix 000660.KS · memory

Rev vs peak Capex ÷ plant dep
At peak 2.17×

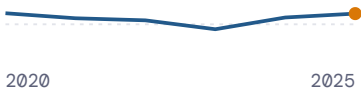
The sharpest recovery in the set — free cash margin from -12.4% to +26.6% in two years. Investment moved just as hard: capital spending fell to 0.65× depreciation in 2023 and is now 2.17×, the highest it has been in the six years.



Samsung 005930.KS · memory

Rev vs peak Capex % of revenue
-3.9% 14.2%

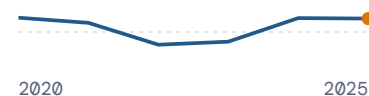
The shallowest cycle of the six. The trough reached only -5.2% where three others passed -12%, and free cash is back to 11.3%. Capital spending is \$33.4bn, 14.2% of revenue — its lowest share of the six years. We do not hold plant depreciation for Samsung, so the replacement ratio is not computed here.



Kioxia 285A.T • memory

Rev vs peak Capex % of revenue
At peak 12.1%

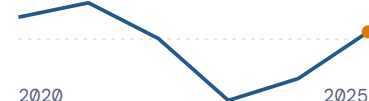
Highest revenue on record, and the least it has spent on plant in six years. Capital spending is 12.1% of revenue against 39.8% in 2022 — half the dollars, in a year revenue reached \$15.6bn. R&D is 24% below its 2021 high. Neither the plant nor the product is being funded out of a record year. We do not hold plant depreciation for Kioxia, so the replacement ratio is not computed here.



Nanya 2408.TW • memory

Rev vs peak Capex ÷ plant dep
-30.3% 0.96×

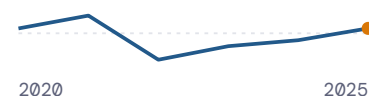
The deepest trough here, and still well below its own peak. Free cash margin reached -64.7% in 2023. Revenue remains 30.3% below peak in the same quarters that three other memory makers set records, and capital spending sits at replacement.



Winbond 2344.TW • memory

Rev vs peak Capex ÷ plant dep
-19.5% 0.53×

Three consecutive negative cash years, and the sharpest reversal in investment shown. Capital spending hit 4.59× depreciation in 2022 — the highest single reading in the set — then fell to 0.53×, about half of replacement. R&D is at a six-year high over the same period, so this reads as a completed build rather than a retreat.



TAKEAWAY

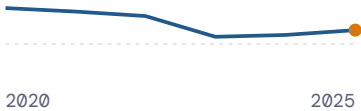
One cycle, two responses. Micron and SK hynix are turning the money into plant at roughly 1.9× and 2.0× depreciation. Kioxia, Nanya and Winbond are at or below replacement — taking the cash rather than building with it. That split, not the price move, is what a fifteen-year part depends on.

Each sparkline is **free cash flow margin** — cash from operations less capital spending, as a share of revenue — for 2020 to the latest full year. One shared scale across all eleven suppliers, -70% to +40%; the dashed line is zero, the marked point is the latest year.

Texas Instruments TXN · analog

Rev vs peak Capex ÷ plant dep
-11.7% **2.37×**

The only supplier here that spent heavily straight through the downturn. Capital spending has been above 2× depreciation for five consecutive years, peaking at 4.32× in 2023, while free cash margin fell from 38.0% to 7.7% before recovering to 14.7%.



STMicroelectronics STM · analog

Rev vs peak Capex ÷ plant dep
-31.7% **1.05×**

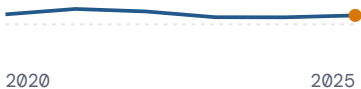
The largest fall from peak of any supplier here, memory included. Revenue is 31.7% below its six-year high and free cash margin fell from 10.9% to 2.6% in two years. Capital spending has come down from 3.17× depreciation in 2022 to 1.05× — a large build, now finished.



Infineon IFX · analog

Rev vs peak Capex ÷ plant dep
-6.2% **1.19×**

Free cash margin halved from 16.2% to 7.4% and has partly recovered to 9.4%. Capital spending has eased from 2.12× depreciation in 2023 to 1.19×, still above replacement. Revenue is close to its peak.



NXP NXPI · analog

Rev vs peak Capex ÷ plant dep
-7.6% **0.71×**

The **steadiest cash generation of the eleven** — between 16.3% and 24.3% in every one of the six years, with no negative period. Capital spending has fallen from 1.93× depreciation in 2022 to 0.71×, the lowest reading of the eleven and below replacement.



Analog Devices ADI · analog

Rev vs peak Capex ÷ plant dep
-10.5% **1.31×**

The **most comfortable position in the set**. Free cash margin never fell below 28.9% and is now 38.8%. Capital spending peaked at 3.76× depreciation in 2023 and is still above replacement at 1.31× — though on a far smaller plant base than the memory makers, so the same ratio represents a fraction of the dollars.



TAKEAWAY

Analog’s spread has almost nothing to do with this cycle — it is the aftermath of the last one. ST built to 3.17× depreciation, stopped, and is now 31.7% below its own peak on 2.6% free cash. Texas Instruments never stopped, and has been above 2× for five years. Same end market, opposite decisions, and the consequences arrive years later.

Nanya and Winbond are both Taiwanese, both below peak and both at or under replacement spending, so they carry the same country and cycle exposure — which matters if they appear on one bill of materials as second sources for one another.

One number here points forward. This is how to read it.

Depreciation is this year's bill for plant already built. Capital spending is what is being built now. **The ratio says which direction the plant base is moving.**

Below 1.0×	Around 1.0×	Above 1.0×
Not replacing what wears out The plant base is shrinking. Whether a specific line survives becomes a real question. Winbond 0.53 NXP 0.71	Replacing what is consumed Steady. Continuity, but no headroom for a step up in demand. Nanya 0.96 · ST 1.05	The plant base is growing Capacity is being added. The roadmap has money behind it. Infineon 1.19 · Analog Devices 1.31 Micron 1.92 · SK hynix 2.17 · TI 2.37

TAKEAWAY

The reading is asymmetric, so trust one end of it and not the other. Below 1.0× is reliable: plant is wearing out faster than it is replaced, and there is no benign reading of that for a fab-owning supplier. Above 1.0× is ambiguous — it can be new capacity, or the same capacity rebuilt on a newer process. The ratio alone cannot tell you which.

A ratio can move for the wrong reason

A fall can mean spending was cut, or that depreciation rose because an earlier build came on line. Identical in the ratio; only the dollars separate them.

It is company-level, never part-level

A supplier can invest heavily overall while ending one legacy device. Nothing in the ratio distinguishes the two.

It is a direction, not a size

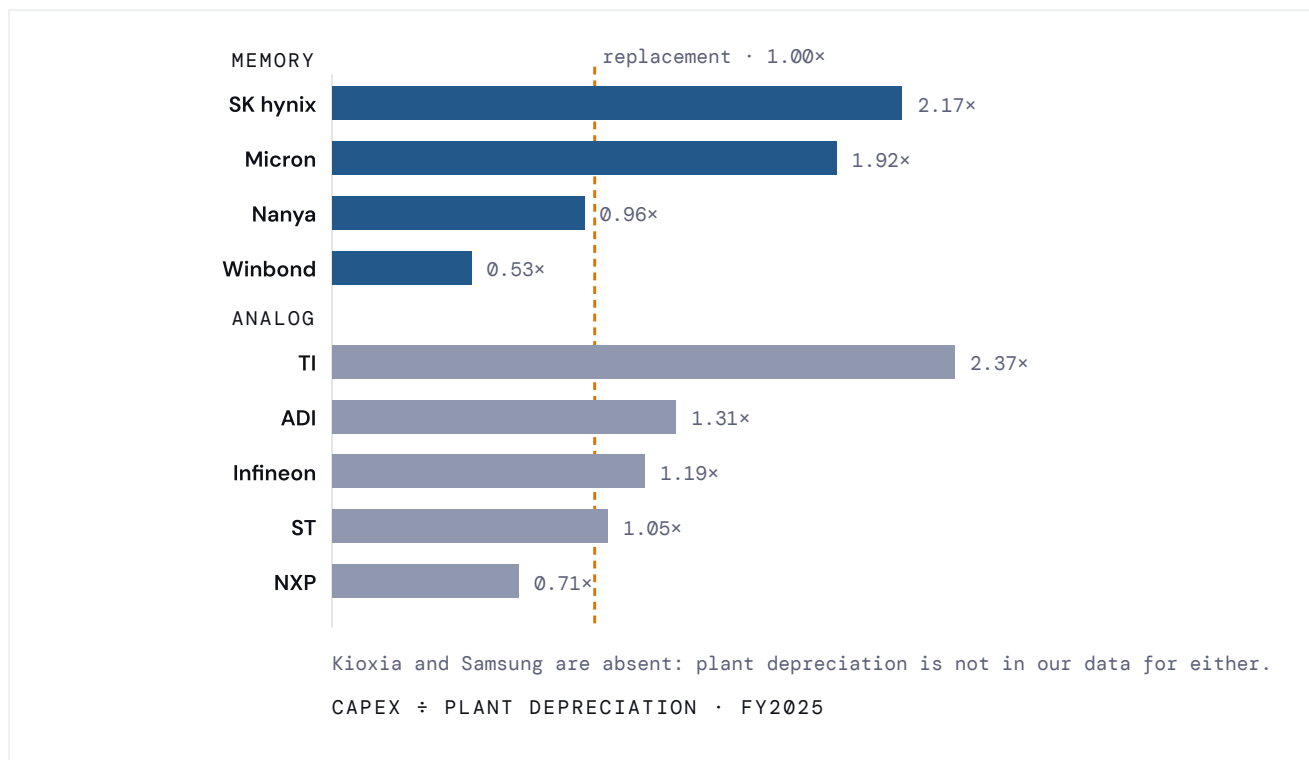
The ratio is scale-free. Analog Devices at 1.31× is spending against \$407m of plant depreciation; Micron at 1.92× against \$8,280m. Both are above replacement; the dollars are not comparable.

Two are missing from this measure

Plant depreciation is not in our data for Kioxia or Samsung, so neither appears in the bands above. Their capital spending is shown as a share of revenue with the memory suppliers instead.

Where the cash from this cycle is going.

The margin expansion in the cycle chart gave the memory suppliers more cash than they had a year earlier; the analog suppliers have been working through a slower cycle of their own. What each is doing with the money differs, and that difference is the closest thing in public data to evidence about whether a part remains available for the life of a vehicle programme.



— Memory · above replacement — Memory · at or below replacement — Analog, for context

Analog Devices reads lowest and is the most comfortable company shown — it owns less plant by design. The ratio is only meaningful within a comparable manufacturing model.

TAKEAWAY

Read this against the price move and it separates two kinds of investment. SK hynix and Micron are building hardest on the largest price moves — capacity underwritten by a cycle. Texas Instruments has spent above 2× for five years on a price that moved ×1.10, and did it through the downturn — capacity underwritten by the business. At the other end, NXP is at its lowest in six years and Winbond at about half of replacement, though Winbond's R&D is at a six-year high — a finished build rather than a retreat. **None of that predicts anything. It says which roadmaps carry money behind them, and under what conditions that money continues.**

One number that explains more supplier behaviour than any roadmap.

Over the six years to the end of the last reported year, Micron reported:

\$157.9bn

of revenue

\$62.2bn

spent on plant and equipment over the same period

\$1.3bn

of free cash generated in total — 0.83% of revenue

This is a snapshot of something that moves

Every figure here has an expiry date, and naming it is the difference between an analysis and a document. The eleven do not report on the same calendar, so the set is never entirely current at one moment.

The specific things that would change the reads in this analysis:

- **Kioxia raising capital spending** back toward its earlier share of revenue — the reading on Kioxia rests on 12.1% against 39.8% in 2022.
- **Winbond returning toward replacement** from 0.53×, or its R&D falling from its six-year high.
- **Nanya regaining its peak**, which closes the divergence inside the memory group.
- **Memory cost of revenue rising with revenue** — the point at which the cycle has turned.

None of those is a prediction — they are the conditions under which these sentences stop being accurate, stated so a reader can check them against the next filings. **A printed page can say where a supplier stands. Only a set that is maintained can say when that stopped being true.**

Six years of work at very large scale, converted into a free cash margin under one per cent, because essentially all of it went back into fabs. That is the economics of memory manufacturing: it consumes capital continuously, and the cycle decides whether the capital comes from earnings or from the balance sheet.

TAKEAWAY

Two things follow. A fab's cost base is largely fixed, so utilisation rather than price is what moves its unit economics — a loaded fab funds the next node, an idle one does not. And a company at this scale on this thin a cash cushion is more exposed to a change in demand than its revenue suggests.

The analog side has its own version of this. ST spent at up to 3.17× depreciation in 2022, and its revenue is now 31.7% below its six-year peak with free cash margin at 2.6% — a build completed into a market that then turned. Texas Instruments has run above 2× for five years, peaking at 4.32×, and is still spending. Same mechanism, different phase of the same decision.

The limits, stated before anyone finds them.

Everything here comes from filed financial statements. That makes it checkable, and blind in specific ways worth being exact about.

It is company-level, not part-level

Nothing in a financial statement identifies a device, a process node or a product line. A supplier can invest heavily overall and still end the specific legacy part in a programme.

It does not see allocation

When supply is tight, who gets the wafers is decided internally by commercial priority. That leaves no trace in a filing until a quarter or more later.

It does not see an end-of-life decision before it is announced

Discontinuation is decided privately and announced on the supplier's schedule. Investment direction can show the pressure behind such a decision; it cannot anticipate one.

It does not see anything negotiated privately

Prices, allocation guarantees, capacity reservations and longevity commitments live in agreements that are not public.

The figures are as filed, and filings lag

Figures are as filed, and filings lag. Where a company reports on a different fiscal calendar its year does not align with the others, so cross-supplier comparison is directional.

TAKEAWAY

What it covers: a supplier's financial position, its investment direction and how it behaved through a full cycle — all from documents anyone can check. **What it is not:** a forecast, a warning system, or a substitute for asking the supplier directly. Its use is in identifying which question is worth asking.

How this was produced.

What this is, in one line: **the economics of semiconductor procurement** — a supplier’s cost structure and capital position, read from documents they published themselves, for the people who have to buy from them.

Sources. Annual and quarterly reports filed by each of the eleven companies — SEC filings for the US filers, their own published reports for the Korean, Japanese, Taiwanese and European ones. Assembling those into one comparable set is the substantive work behind this document.

Only a minority of the eleven file a US annual report; the rest publish under their own national regimes. That is why cross-supplier comparison of this kind is uncommon — not because the documents are secret, but because they are inconsistent.

Definitions used throughout.

TERM	DEFINITION AS USED HERE
Free cash margin	Cash from operations less capital expenditure, divided by revenue.
Capital intensity	Capital expenditure divided by depreciation of property, plant and equipment — plant depreciation only, never total depreciation and amortisation , which for some companies differs by a factor of several. Not computed for Kioxia or Samsung, whose plant depreciation we do not hold.

Semitrical builds supplier intelligence for semiconductor procurement — what a chip supplier's own filings say about where they stand, before you negotiate. This is one snapshot of that analysis, covering the memory and analog suppliers behind a vehicle programme. It may be forwarded and shared freely.

Questions, corrections and disagreements are all welcome — a correction from someone who knows a supplier better than a filing does is worth more than the page it changes.

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